

E-Commerce & Retail Store Analysis

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Tools: Python (EDA) | Power BI
(Dashboard)

Welcome to a Data Driven Insight

Unlocking growth and optimizing Performance In an Ecommerce Analytics

Company Background

Brief overview of the company:

An e-commerce company is facing multiple operational challenges that hinder growth and customer satisfaction. To address these issues, the company is leveraging data-driven insights to optimize operations.

Key business areas: financial performance, customer insights, logistics & fulfillment, product management, customer satisfaction & returns

Dataset covers **4 months of sales data**

FINANCIAL PERFORMANCE ANALYSIS

Analyzing total revenue growth over time. Identifying best-selling products and seasonal sales trends. Calculating the average order value to understand spending behavior

CUSTOMER INSIGHTS ANALYSIS

Identifying top-performing locations based on sales volume. Analyzing order cancellation trends and customer retention. Differentiating B2B vs. B2C customer behavior for targeted marketing

LOGISTICS & FULFILLMENT OPTIMIZATION

Addressing shipping delays by calculating average shipping times. Comparing orders fulfilled by Amazon vs. Merchant fulfillment. Improving order processing for faster deliveries

PRODUCT MANAGEMENT & INVENTORY OPTIMIZATION

Identifying high-demand product categories. Analyzing the average quantity ordered for efficient stock management. Enhancing inventory control to prevent stockouts or overstocking

CUSTOMER SATISFACTION & RETURNS

Investigating high return and cancellation rates. Identifying problematic products/categories to enhance quality. Improving customer experience through better service policies

Problem Statement

Orders are being delayed, canceled, or returned frequently, affecting customer satisfaction and revenue. The company wants to identify operational inefficiencies and optimize product and fulfillment performance.

Challenges Faced

- Data quality issues (missing or inconsistent order statuses)
 - Date hierarchies for trend analysis (month, week, weekday)
 - Identifying operational KPIs in a concise one-page dashboard
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E-commerce & Retail Store Dashboard

Month

All

Category

All

Total Revenue

71.79M

+42.53% Revenue growth

128,943

Count of Orders

596.50

AOV

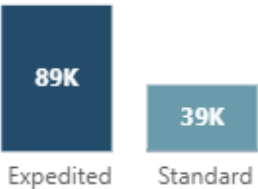
22.31%

OnTime Delivery Rate (%)

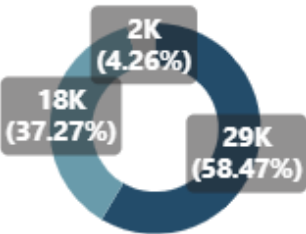
14.22%

Cancellation Rate

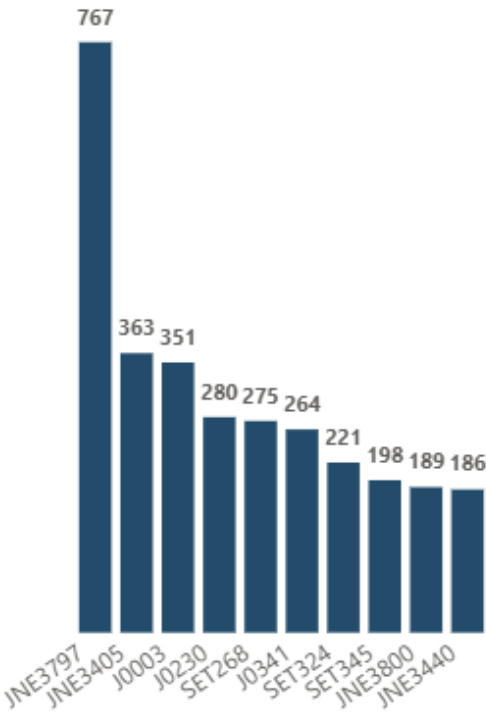
Delivery Performance



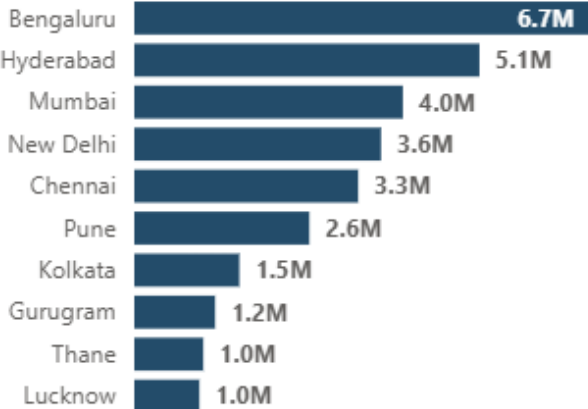
Returns & Cancellations Overview



Top Problematic Products



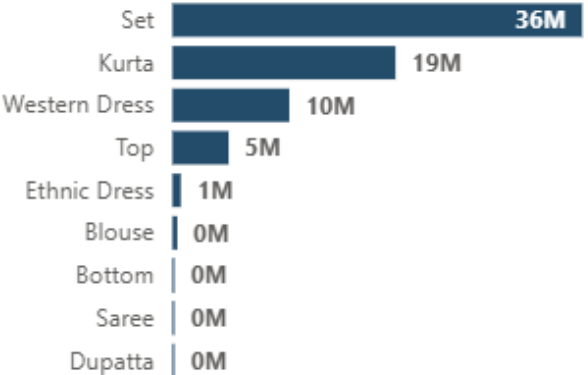
Top Performing City



Revenue by Sales Channels

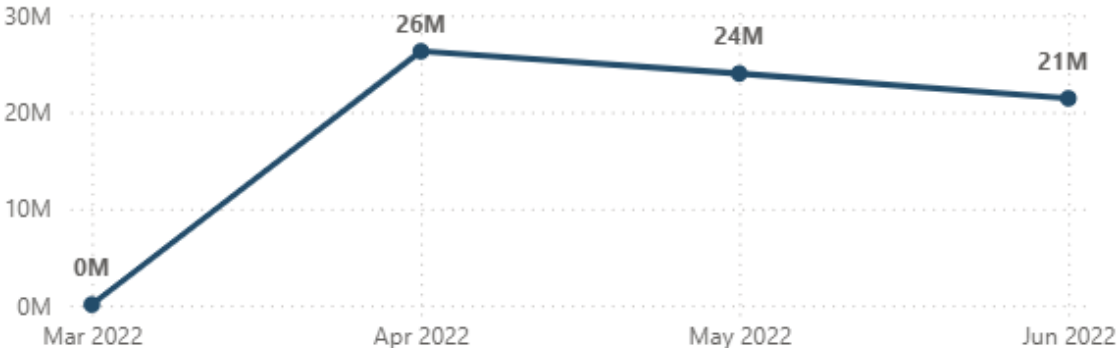


Revenue by category



Monthly Weekly Daily

Monthly Revenue Trend



Key Performance Indicators

**Total Revenue:
₹71.79M & Revenue
Growth (MoM):
42.53%**

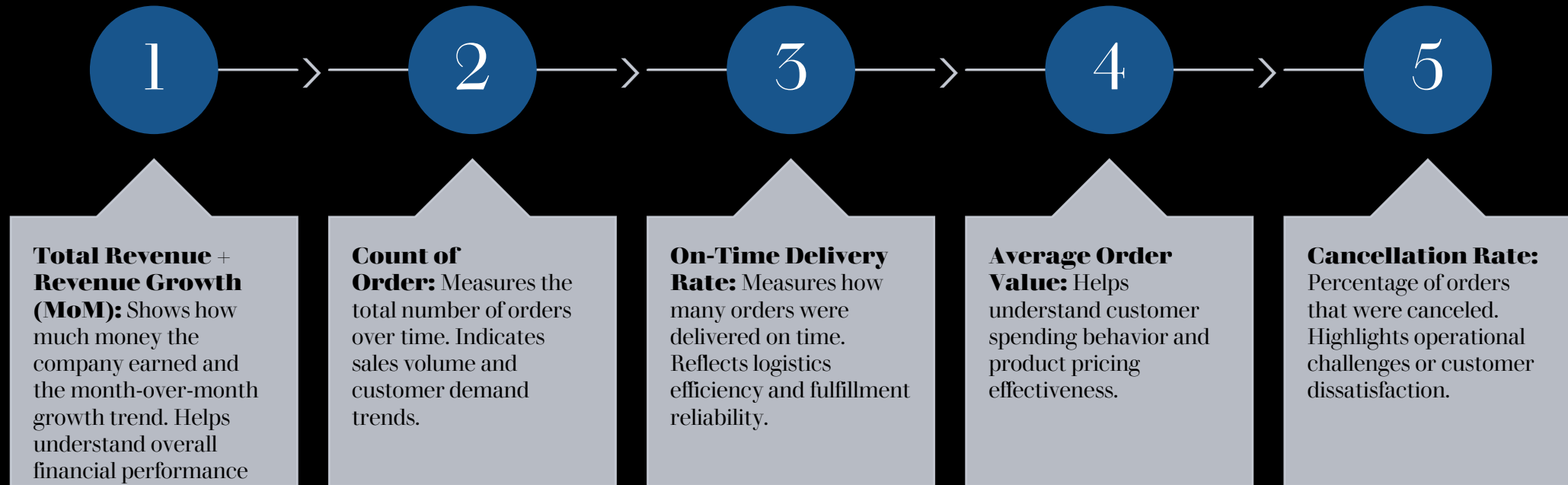
**Count of
Order: 128,943**

**On-Time
Delivery Rate:
22.31%**

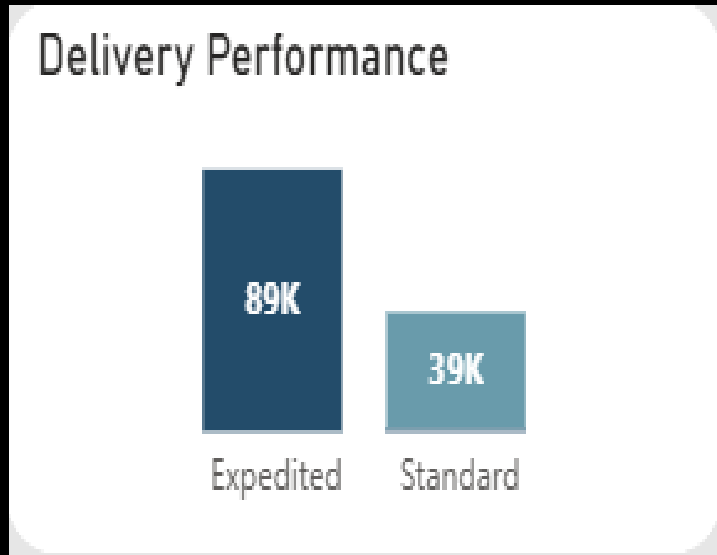
**Average Order
Value: ₹596.50**

**Cancellation
Rate: 14.22%**

KPI Explanation



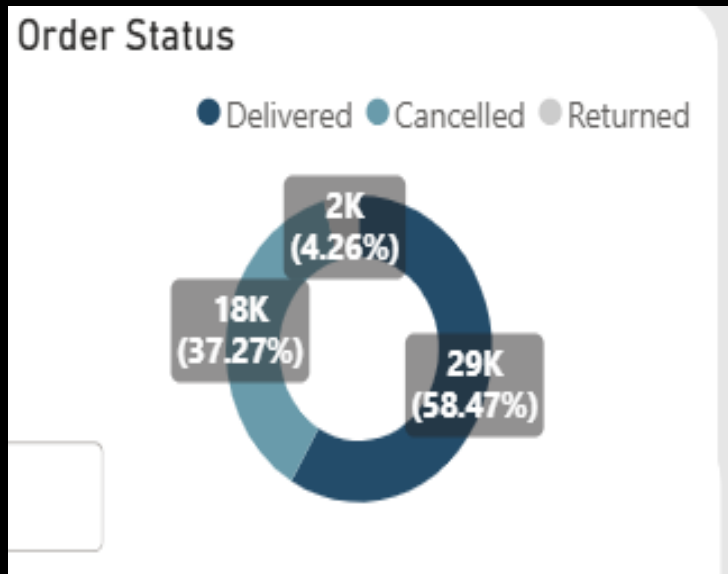
Delivery Performance



Purpose of the Chart: Shows how delivery speed and fulfillment method impact delivery performance. Helps identify which service level and fulfillment type achieve higher on-time delivery.

Key Insight: Expedited shipping fulfilled by Amazon has significantly higher delivery numbers compared to standard shipping, indicating that faster shipping methods and reliable fulfillment partners improve operational efficiency.

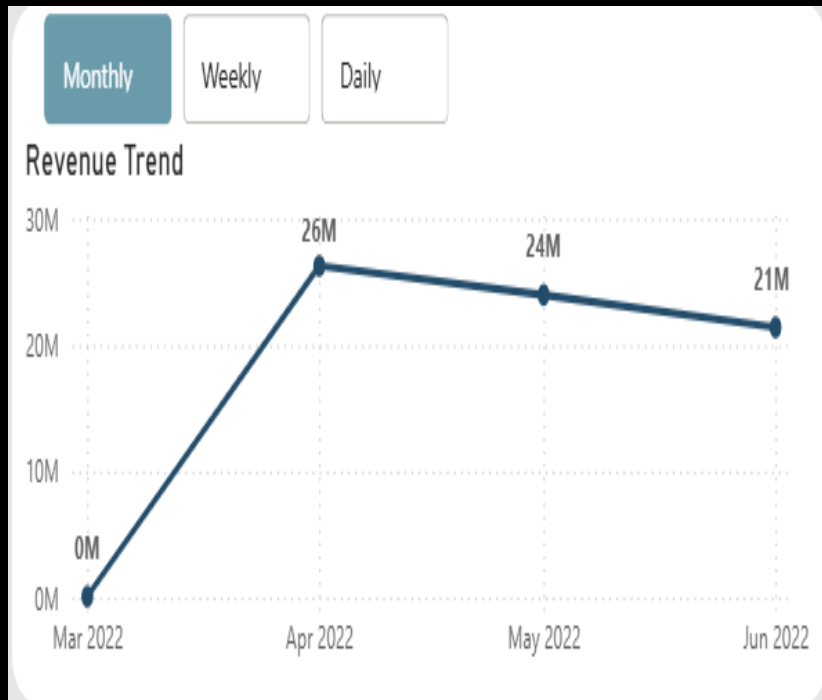
Order Status



Purpose of the Chart: Provides a quick snapshot of how orders are completing versus failing. Highlights customer satisfaction and operational efficiency.

Key Insight: Most orders (58.47%) were successfully delivered. However, a high percentage of canceled orders (37.27%) indicates operational challenges or customer dissatisfaction. Returned orders are relatively low (4.26%) but still point to product or fulfillment issues that need attention.

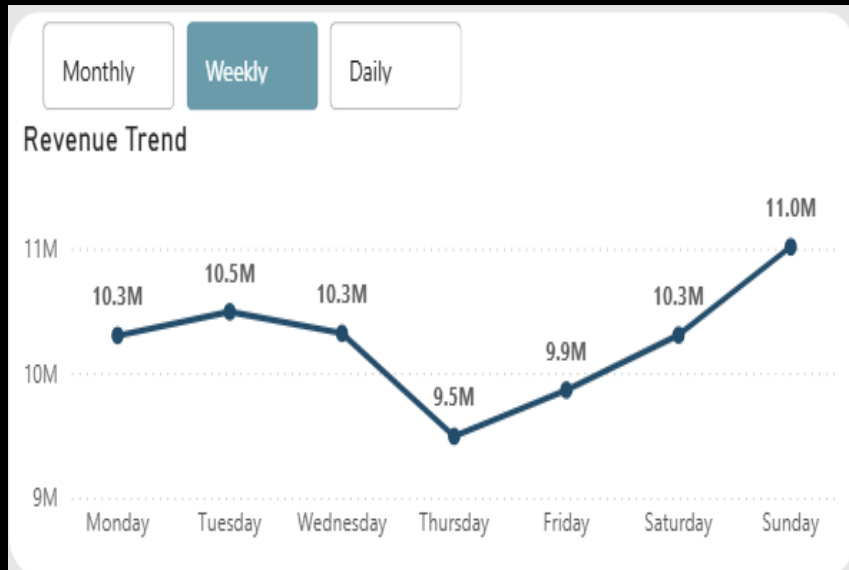
Monthly Revenue Trend



Purpose of the Chart: Shows how revenue changes across each month within the analysis period.

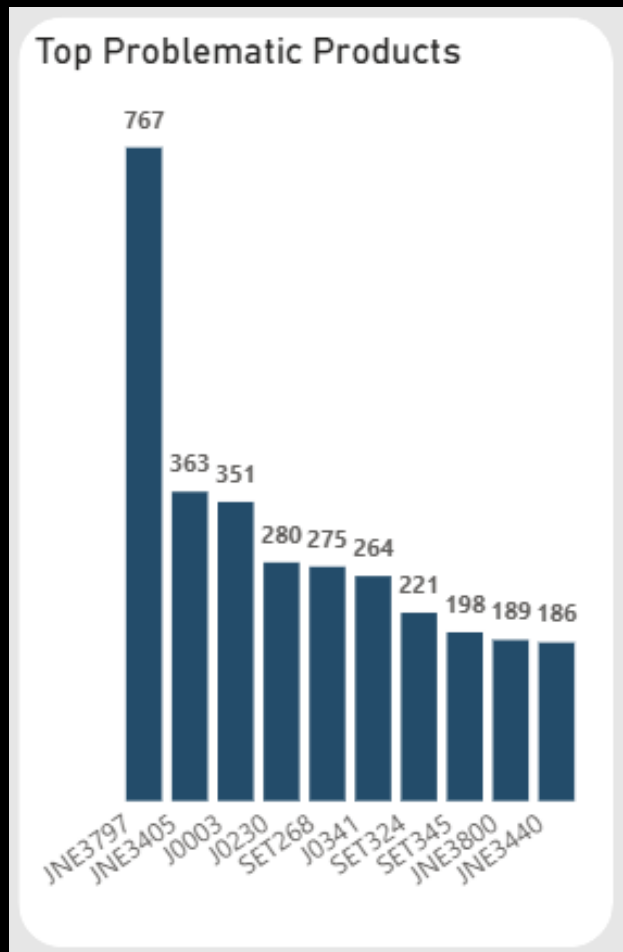
Key Insight: This chart shows how much money the company made each month. Revenue increased sharply in April, then reduced slightly in May and June, showing where growth happened and where improvement is needed.

Weekly Revenue Trend



Purpose of the Chart: Analyzes customer purchasing behavior across weekdays to identify high-performing and low-performing days.

Key Insight: Revenue is highest on Sundays and early in the week. Sales drop toward Thursday and Friday, showing when customers are most and least active.

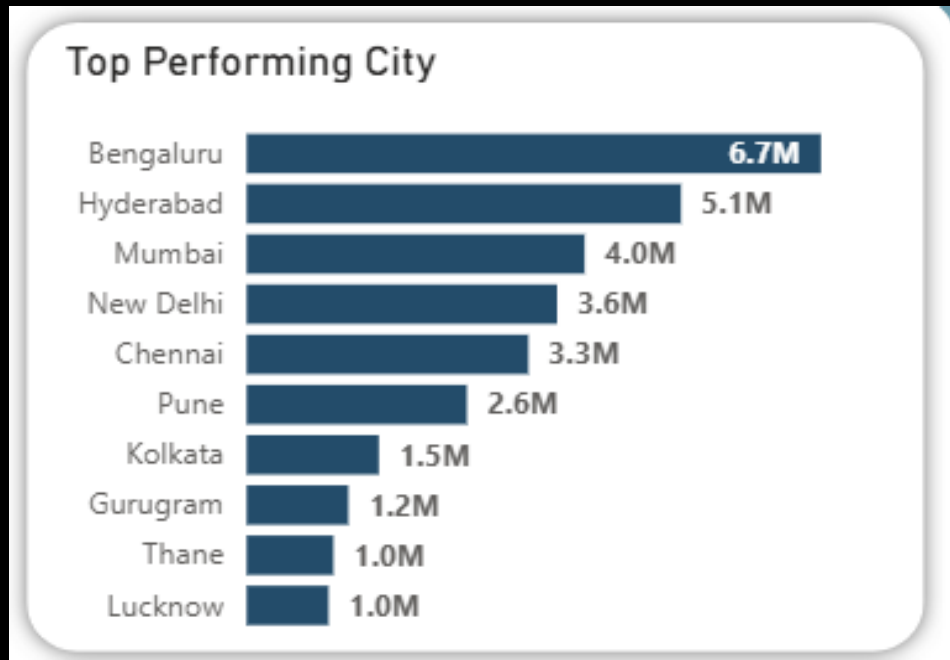


Top Problematic Products

Purpose of the Chart: Identifies product styles with the highest number of failed orders to pinpoint operational and customer satisfaction issues.

Key Insight: A small number of product styles account for a large share of failed orders, with JNE3797 standing out as the biggest operational risk. The high concentration of failed orders around specific product styles suggests potential issues related to **product quality, sizing, fulfillment accuracy, or customer expectations**.

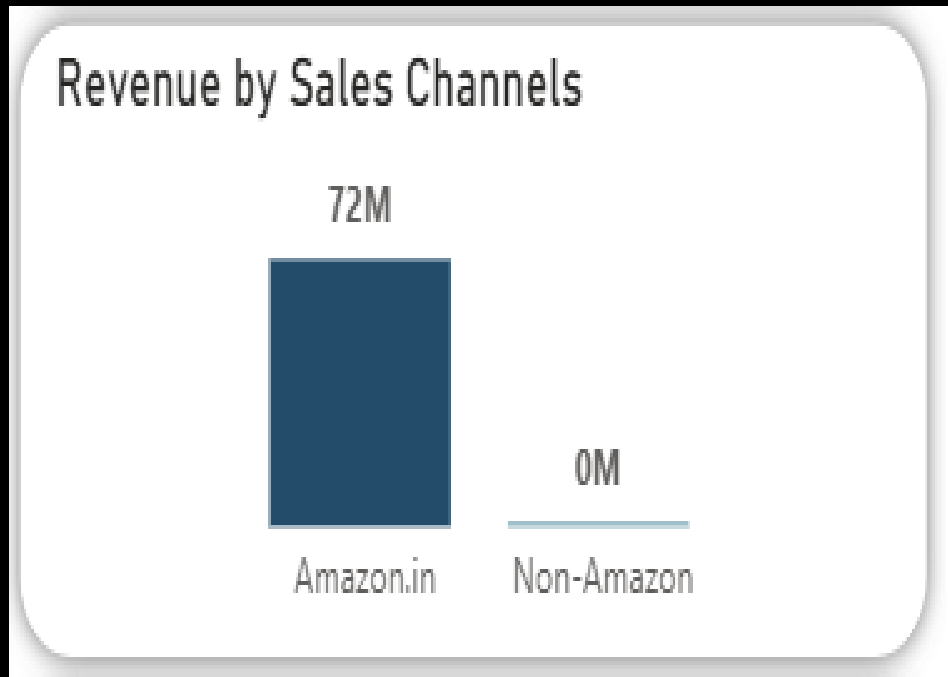
Top Performing City by Revenue



Purpose of the Chart: To identify the highest-revenue-generating cities and understand where the business is currently strongest geographically.

Key Insight: Bengaluru, Hyderabad, and Mumbai dominate revenue among the top 10 cities.

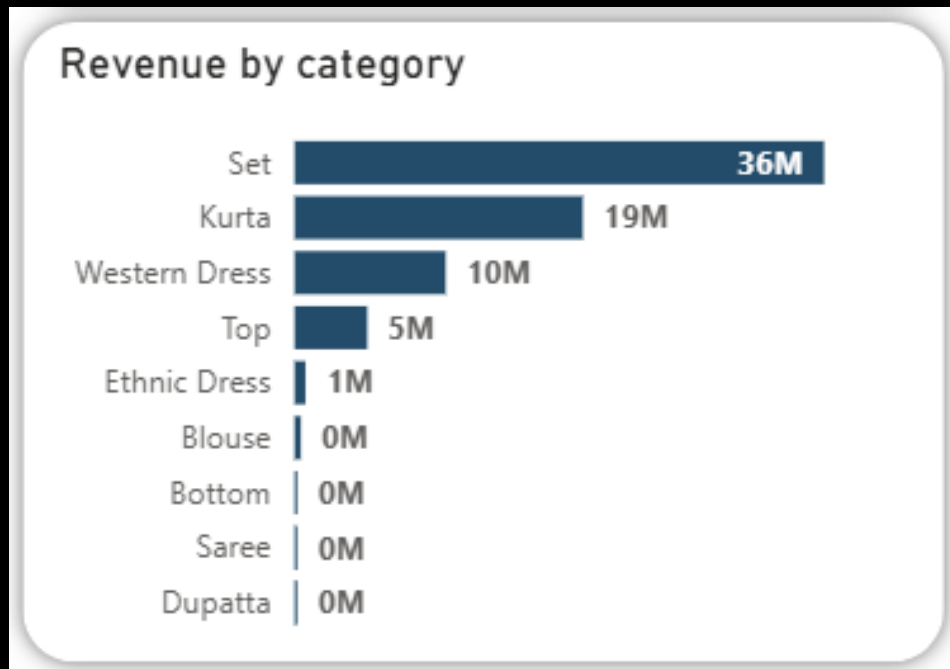
Revenue by Sales Channels



Purpose of the Chart: To evaluate how revenue is distributed across sales channels and assess the company's dependence on each channel.

Key Insight: While Amazon.in is a strong revenue driver, the low contribution from non-Amazon channels highlights a **potential risk of channel dependency** and an **opportunity to diversify revenue sources**.

Revenue by Category



Purpose of the Chart: To identify which product categories drive the most revenue and understand customer purchase preferences.

Key Insight: Revenue is highly concentrated in a few categories, indicating that **Sets, Kurtas and western dress** are the company's core revenue drivers, while several categories underperform and may require repositioning, promotion, or rationalization.

Conclusion

The analysis of the e-commerce and retail store data highlights several key insights:

- **Revenue Performance:** Strong growth was observed in April, with total revenue driven primarily by Amazon.in and top categories such as Sets, Kurtas, and Western Dresses.
- **Customer & Operational Insights:** Delivered orders constitute 58.47% of total orders, but a high cancellation rate (37.27%) indicates operational or fulfillment challenges.
- **Top Performing Markets:** Bengaluru, Hyderabad, and Mumbai are the leading revenue-generating cities.
- **Problematic Products:** Specific product styles, notably JNE3797, show disproportionately high failed orders, signaling product or fulfillment issues.
- **Channel & Category Dependence:** Heavy reliance on Amazon.in and a few key categories highlights potential risk concentration.

Overall, the company exhibits strong growth potential but faces operational inefficiencies and revenue concentration risks that require strategic action.

Recommendation

1. **Operational Improvements:** Investigate high cancellation and return rates; enhance fulfillment accuracy and delivery reliability.
 2. **Product Focus:** Review top problematic products (e.g., JNE3797) for quality, sizing, or listing improvements to reduce failures.
 3. **Channel Diversification:** Strengthen non-Amazon sales channels to reduce dependency and mitigate risk.
 4. **Market Strategy:** Prioritize marketing and inventory for high-performing cities (Bengaluru, Hyderabad, Mumbai) while exploring growth opportunities in underperforming cities.
 5. **Category Optimization:** Focus on high-revenue categories (Sets, Kurtas, Western Dresses) while reassessing low-performing categories for promotion, repositioning, or discontinuation.
 6. **Data-Driven Monitoring:** Implement continuous dashboard monitoring to track revenue trends, returns, cancellations, and top-performing products for proactive decision-making.
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Thank You

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